MIL-E-1/1394B(USAF)
7 JAN 1977
SUPERSEDING
MIL-E-1/1394A(USAF)
14 November 1972

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NOISE SOURCE

TYPE 6357

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: X-Band, gaseous discharge diode

ABSOLUTE RATINGS:

Parameter:	I_f	TA	T_{bulb}
Unit:	mAdc	°C	°C
Maximum:	170	+85	+125
Minimum:	---	-54	---

PHYSICAL CHARACTERISTICS:

Cathode:	Filamentary type
Dimensions:	See outline drawing
Base:	See outline drawing
Mounting position:	Any

TEST CONDITION:

Parameter:	I_f	I_b
Unit:	A	mAdc
Value:	0	200

GENERAL:

Preproduction Test - Required (See note 10)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1					
1261	Filament voltage	I_f = 170 mAdc	E_f	---	20	Vdc
1337	Voltage drop	See note 2	Etd	70	80	Vdc
----	Excess noise	F = 9.0 ±0.3% GHz See notes 1, 3, 4 and 5	NR-1	15.40	15.80	db
----	Match (1)	F = 9.0 ±0.3% GHz Ib = 0 See notes 4 and 6	VSWR	---	1.1:1	---
----	Match (2)	F = 9.0 ±0.3% GHz Ib = 250 mAdc See notes 4 and 6	VSWR	---	1.1:1	---
	Quality conformance inspection, part 2					
1031	Variable frequency vibration	No voltages See note 9	---	---	---	---
	Quality conformance inspection, part 3					
----	Intermittent life test	Group D; Cycle one minute on, 2 minutes off; Preheat time 2 to 3 seconds See notes 3 and 8	---	2500	---	Cycles
	Life test end point:					
----	Excess noise ratio	F = 9.0 ±0.3% GHz See notes 1, 3, 4 and 5	Nr-1	15.35	15.85	db

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.

NOTES:

1. The excess noise ratio (Nr-1) is defined in db as $Nr-1 = 10 \log \left(\frac{Te}{290}-1\right)$ where Te is the effective electron temperature.
2. In the test circuit of Figure 4 with a filament current of 170 mAdc, the tube shall operate within three tries.
3. The tube shall be tested in total darkness.
4. The tube shall be tested in a tube mount as specified in Figure 3, or equivalent, terminated by a matching RG-52/U termination having a VSWR no greater than 1.01:1 such as a Hewlett-Packard X-914A, or equal. Excess noise ratio measurements tests shall be made using the circuit of the block diagram of Figure 2. Excess noise ratio should be measured by comparison with an approved standard noise source.
5. The frequency specified is that of the local oscillator.
6. The frequency specified is that of a signal generator.
7. Excess noise ratio shall be measured by comparison with an approved standard.
8. The tube shall be tested at an ambient temperature of plus 85°C (185°F).
9. Intermittent life test end points shall apply.
10. First article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified on this specification sheet. In addition, first article sample approval shall include satisfactory demonstration of tube-to-system compatibility. Invitation for bids should provide that the preparing activity reserves right to waive the requirements for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.
11. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level of II.

Custodian:
Air Force - 85

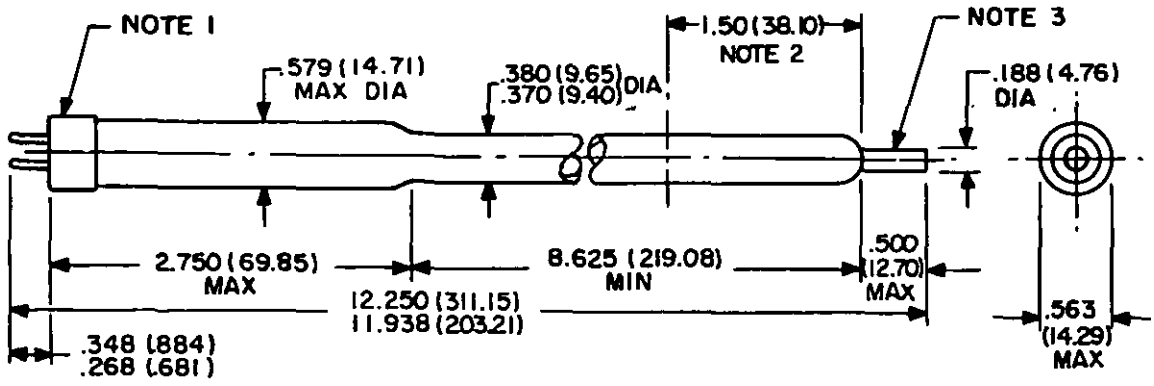
Preparing activity:
Air Force - 85

Review activities:
Air Force - 99

(Project No. 5960-F075)

User activities:
Air Force - 11

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.



NOTES:

1) MINIATURE BI-PIN BASE.
2) .380 MAX., NO MIN DIMENSION ON DIAMETER FOR THIS LENGTH OF TUBE.
3) BRASS CONTACT.
4) DIMENSIONS SHOWN IN INCHES. MILLIMETERS SHOWN IN PARENTHESES.

FIGURE 1 Outline drawing for type 6357

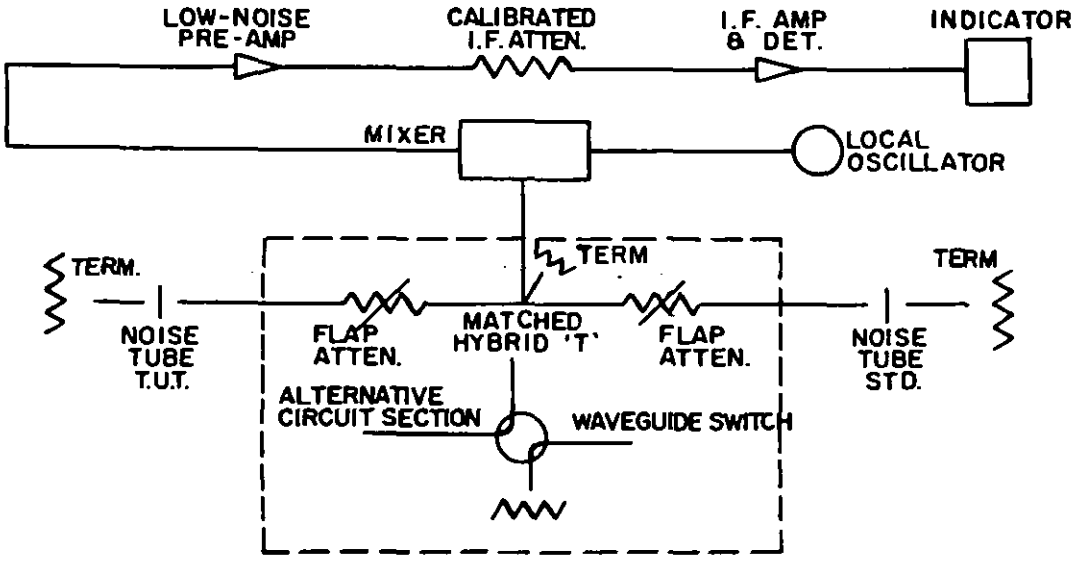


FIGURE 2 Test circuit for excess noise measurements

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.

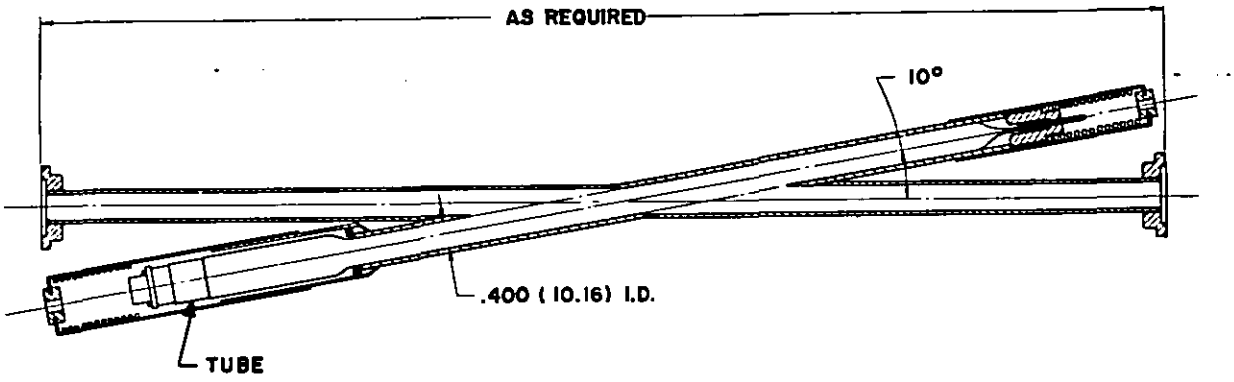


FIGURE 3 Waveguide mount assembly RG-52/U

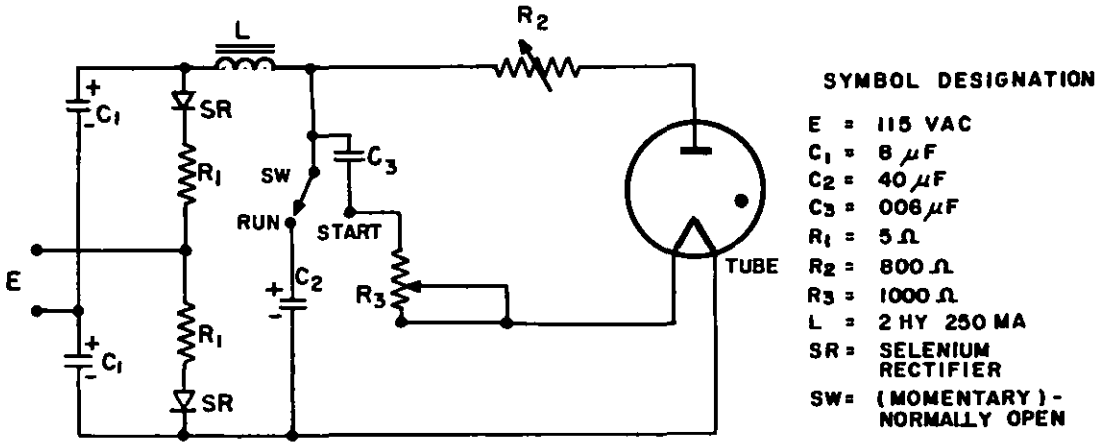


FIGURE 4 Starting and operating circuit

☆ U.S. GOVERNMENT PRINTING OFFICE: 1977—703-020/1346

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A.. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

☆ U.S. GOVERNMENT PRINTING OFFICE: 1972—714-863/3404 2-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:38Z
Check the source to verify that this is the current version before use.